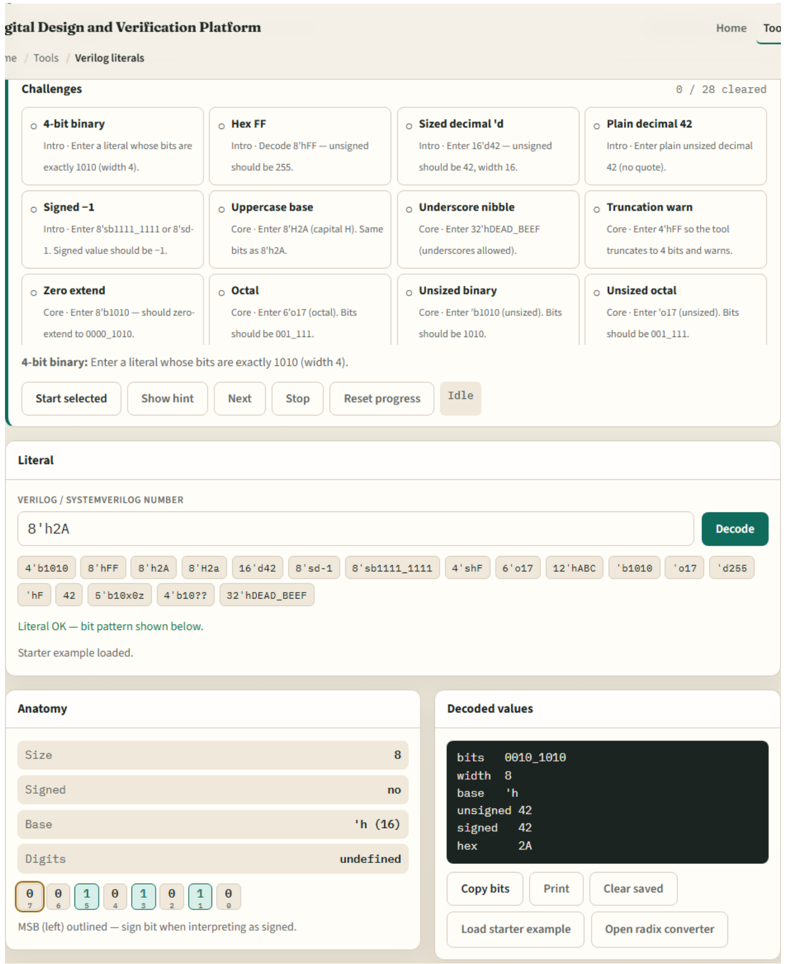
Verilog literals

Constants show up everywhere in RTL, reset values, parameter defaults, and compare masks

How literals are built

- A sized literal has three parts
- Eight apostrophe h F F is eight bits of all ones
- An unsized literal like apostrophe b one zero one zero still has a base but lets the tool
- Add s after the apostrophe for signed interpretation
- Underscores between nibbles are cosmetic; they do not change the value

Browser lab



Verilog literals lab starter

Real Verilog practice

```
wsl — module02-verilog-literals/examples/navigation

# Track A — real Unix

# real Verilog session (Track A)

$ cat examples/literals/literals.v
// literals.v — sized bases for Track A practice
module literal_demo;
  localparam [7:0]  MASK_FF = 8'hFF;
  localparam [3:0]  PATTERN = 4'b1010;
  localparam [15:0] COUNT   = 16'd42;
  localparam [7:0]  NEG1    = 8'shFF;
endmodule

$ iverilog -t null examples/literals/literals.v
(syntax check passed)
```

Real Verilog — literal localparams

Pitfalls to watch

- Do not ignore width, four apostrophe h F F truncates to four bits
- Do not treat signed and unsigned as interchangeable
- X and Z are for simulation and unknown states
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, load the starter and finish a few challenges
- On real Verilog, write or edit a minimal sketch with at least two different bases
- When you are ready